

States Of Matter

1. A gas mixture containing 160 parts per million by volume of H_2S in Ne has a total pressure of 725 mmHg. What is the partial pressure of H_2S ?

(A) 0.069 mmHg (B) 0.12 mmHg
(C) 0.16 mmHg (D) 0.20 mmHg

2. Which gas has the highest root-mean-square velocity at 298 K?

(A) CO (B) CO_2 (C) CH_4 (D) CF_4

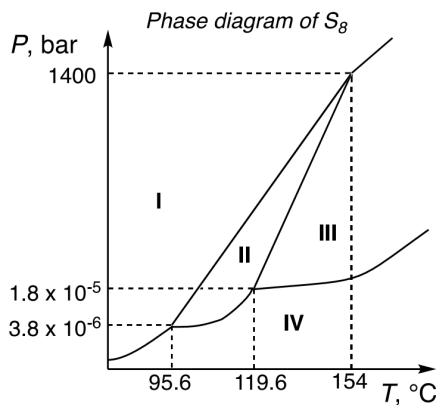
3. Which statement describes a change when a pure solid melts spontaneously?

(A) The entropy decreases.
(B) The free energy increases.
(C) The vapor pressure increases.
(D) The process is endothermic.

4. Which compound has the highest vapor pressure at 25 °C?

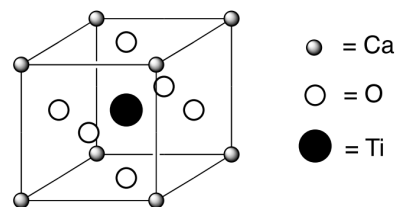
(A) $\text{CH}_3\text{CH}_2\text{CH}_2\text{OH}$ (B) $\text{CH}_3\text{CH}_2\text{OCH}_3$
(C) $\text{CH}_3\text{CHOHCH}_3$ (D) $\text{HOCH}_2\text{CH}_2\text{OH}$

5. Sulfur has two stable solid phases, monoclinic and orthorhombic sulfur. Which regions of the phase diagram correspond to these two solid phases?



(A) I and II (B) I and IV
(C) II and III (D) II and IV

6. The mineral perovskite has the cubic unit cell shown below, with the oxygens occupying the centers of the faces, the titanium in the center, and the calciums on the corners of the cube. What is the formula of perovskite?



(A) CaTiO_3 (B) Ca_2TiO_4
(C) CaTi_2O_4 (D) Ca_8TiO_6

7. 1.53 g of a gas occupies a volume of 264 mL at 722 mm Hg and 22 °C. What is the molecular formula of the gas?

(A) C_2F_6 (B) C_3HF_5
(C) $\text{C}_3\text{HF}_5\text{O}$ (D) $\text{C}_3\text{F}_6\text{O}$

8. A mole of which gas will have a volume closest to 22.4 L at 0 °C and 1 atm pressure?

(A) H_2 (B) CO_2
(C) SF_6 (D) Xe

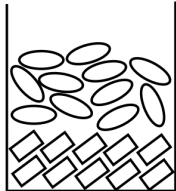
9. Two identical steel cylinders at the same temperature contain equal masses of CO(g) and $\text{C}_2\text{H}_2\text{(g)}$. Which quantities will be greater for the cylinder containing C_2H_2 ?

I. Pressure II. Average molecular velocity
(A) I only (B) II only
(C) Both I and II (D) Neither I nor II

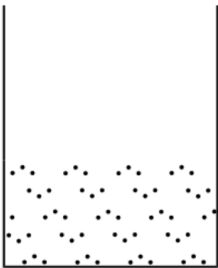
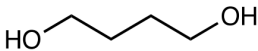
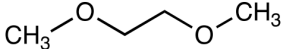
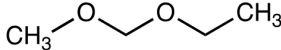
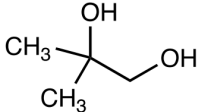
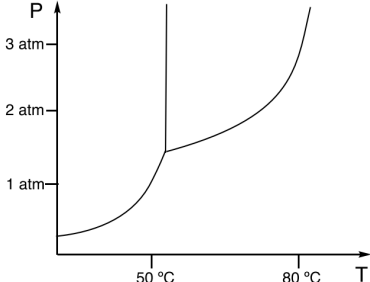
10. The critical point of carbon dioxide is 31.0 °C and 72.8 atm. Which best describes the composition of a sample of pure CO_2 at 29.0 °C and 75 atm?

(A) Only gaseous CO_2 is present.
(B) Only liquid CO_2 is present.
(C) A mixture of gaseous CO_2 and liquid CO_2 is present.
(D) Only supercritical CO_2 is present.

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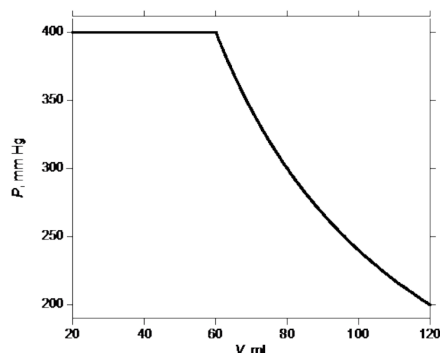
11. The normal boiling point of ethane (C_2H_6) is -88.5°C , while the normal boiling point of fluoromethane (CH_3F) is -78.4°C . A difference in which type of intermolecular forces is most responsible for the higher boiling point of fluoromethane?
- (A) London dispersion forces
(B) Dipole-dipole interactions
(C) Ion-dipole interactions
(D) Hydrogen bonding
12. Cadmium metal crystallizes in a hexagonal closest packed structure. How many nearest neighbors does a cadmium atom have in this structure?
- (A) 4 (B) 6 (C) 8 (D) 12
13. A white crystalline solid melts at 400°C and dissolves in water to give an electrically conducting solution. Which is it?
- (A) LiOH (B) $\text{C}_{12}\text{H}_{22}\text{O}_{11}$
(C) SiO_2 (D) Cu
14. A 10.0 L vessel containing 5.0 atm of He and a 3.0 L vessel containing 10.0 atm of Ar are connected by a valve. The valve is opened and the gases are allowed to mix at constant temperature. What is the final pressure in the vessels?
- (A) 2.3 atm (B) 3.8 atm
(C) 6.2 atm (D) 7.5 atm
15. A mixture of **A** (whose molecules are represented by rectangles) and **B** (whose molecules are represented by ovals) is represented schematically below. Which is the best description of this mixture?
- 
- (A) **A** is dissolved in liquid **B**.
(B) **A** is a crystalline solid in the presence of liquid **B**.
(C) **A** and **B** are crystalline solids with **A** denser than **B**.
(D) **A** and **B** are immiscible liquids with **A** denser than **B**.
16. Which statement best describes a sample of ammonia at equilibrium at its triple point (195.4 K , 0.0606 bar)?
- (A) Gaseous ammonia must be present, and solid or liquid ammonia may be present.
(B) Liquid ammonia must be present, and solid or gaseous ammonia may be present.
(C) Solid ammonia must be present, and liquid or gaseous ammonia may be present.
(D) Gaseous, liquid, and solid ammonia must all be present.
17. Which compound's lattice energy is largest in magnitude?
- (A) LiF (B) NaCl (C) MgO (D) CaS
18. Tantalum crystallizes in a body-centered cubic structure. How many nearest neighbors does each tantalum atom have?
- (A) 4 (B) 6 (C) 8 (D) 12
19. A room temperature sample of xenon gas in a metal container with a piston has a volume of 1.00 L and a pressure of 4.0 atm. The piston is moved to change the volume of the container to 2.00 L, after which the pressure reads 1.75 atm. Which is the best explanation for this observation?
- (A) The volume is miscalibrated and the actual final volume is 1.75 L.
(B) Some xenon has leaked out through the seal in the piston.
(C) The friction of moving the piston has caused the temperature of the gas to increase.
(D) The system is correctly calibrated, well sealed, and remains at room temperature.
20. In which are the root mean square velocities of Xe(g) , Ne(g) , and $\text{N}_2\text{(g)}$ at 290 K and 3.0 atm ranked from lowest to highest?
- (A) $\text{Xe} < \text{Ne} < \text{N}_2$ (B) $\text{Xe} < \text{N}_2 < \text{Ne}$
(C) $\text{N}_2 < \text{Ne} < \text{Xe}$ (D) $\text{Ne} < \text{N}_2 < \text{Xe}$
21. In a sample of pure silane (SiH_4), which intermolecular forces are present?
- I. London dispersion forces
II. Hydrogen bonding
- (A) I only (B) II only
(C) Both I and II (D) Neither I nor II

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22. The melting point of an organic compound X is $5.0\text{ }^{\circ}\text{C}$. How do the vapor pressures of $X(s)$ and $X(l)$ compare at $10.0\text{ }^{\circ}\text{C}$?
- (A) The vapor pressure of $X(s)$ is less than the vapor pressure of $X(l)$.
(B) The vapor pressure of $X(s)$ is equal to the vapor pressure of $X(l)$.
(C) The vapor pressure of $X(s)$ is greater than the vapor pressure of $X(l)$.
(D) The relative vapor pressures cannot be determined without knowledge of the relative densities of $X(s)$ and $X(l)$.
23. The arrangement of atoms (represented as black dots) in a pure substance contained in a beaker is shown schematically below. Which is the best description of the phase of the substance?
- 
- (A) Crystalline molecular solid
(B) Crystalline network covalent solid
(C) Amorphous network covalent solid
(D) Liquid
24. Which statement about a sample of ammonia (NH_3) at its triple point is correct?
- (A) It has a pressure of 1.0 atm .
(B) It can be at only one possible temperature.
(C) There are equal amounts of NH_3 in the solid, liquid, and gas phases.
(D) If it is in a closed, well-insulated container, it will remain at the triple point when the volume of the container is decreased by a factor of two.
25. What is the partial pressure of Ar in a gaseous mixture of 3.25 g Ar and 2.28 g Kr in a 5.00 L container at $15.0\text{ }^{\circ}\text{C}$?
- (A) 0.0572 bar (B) 0.390 bar
(C) 0.412 bar (D) 0.447 bar
26. Which isomer of $\text{C}_4\text{H}_{10}\text{O}_2$ has the highest normal boiling point?
- (A) 
(B) 
(C) 
(D) 
27. Which statements about the behavior of gaseous H_2 molecules in a container at 1 atm and 298 K are correct?
- I. All H_2 molecules are moving at the same speed.
II. The H_2 molecules are colliding more frequently with the walls of the container than they would in the same container at 398 K .
- (A) I only (B) II only
(C) Both I and II (D) Neither I nor II
28. A portion of the phase diagram of uranium hexafluoride (UF_6) is shown below.
- 
- Which statements are correct?
- I. UF_6 sublimes, rather than melts, at 1 atm pressure.
II. At 2 atm and $45\text{ }^{\circ}\text{C}$, only solid UF_6 is present at equilibrium.
- (A) I only (B) II only
(C) Both I and II (D) Neither I nor II
29. How many nearest neighbors does each silicon atom have in a crystal of Si?
- (A) 4 (B) 6
(C) 8 (D) 12

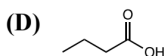
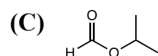
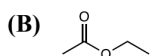
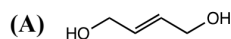
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30. A sample of methanol, CH_3OH , is introduced into an evacuated chamber with a movable piston. The pressure is measured as a function of the volume of the container while the temperature is maintained at 50°C , and the graph below is obtained. Which statement best describes this system?



- (A) At volumes below 60 mL, methanol is only present in the liquid phase.
 (B) At volumes above 60 mL, methanol is only present in the gaseous phase.
 (C) If the process were repeated at 60°C , the point at which the pressure begins to decrease would shift to higher volume.
 (D) If more methanol were added to the container, the pressure at $V = 20\text{ mL}$ would increase.
31. When liquid water converts to solid ice at 0°C and 1 atm pressure, which of the following changes takes place?
- (A) Heat flows out of the system.
 (B) The volume of the system decreases.
 (C) The vapor pressure of water increases.
 (D) The hydrogen bonds become weaker.

32. Which isomer of $\text{C}_4\text{H}_8\text{O}_2$ has the greatest solubility in water?

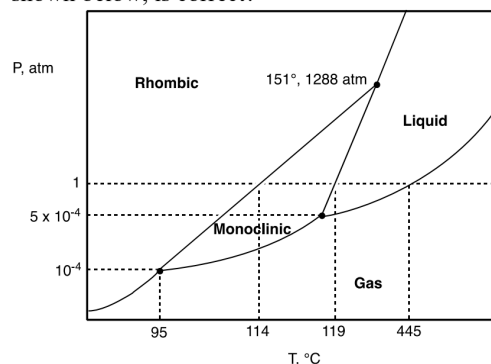


33. Which compound has the lowest normal boiling point?
- (A) HF (B) HCl (C) HBr (D) HI

34. A sealed container contains a gaseous sample at 300.0 K consisting of either pure ethane (C_2H_6), pure neon (Ne), or a mixture of the two. The pressure inside the container at this temperature is 1.00 atm. When the container is cooled to 150.0 K , the pressure is 0.37 atm. What is the composition of the sample? (The vapor pressure of C_2H_6 at 150 K is 0.10 atm; the critical temperature of Ne is 44 K .)

- (A) Pure ethane
 (B) Pure neon
 (C) A mixture of ethane and neon.
 (D) It cannot be determined from the information given.

35. Which statement about sulfur, whose phase diagram is shown below, is correct?

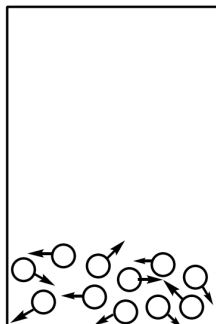


- (A) Monoclinic sulfur sublimates at 1 atm pressure while rhombic sulfur does not.
 (B) Rhombic sulfur is less dense than monoclinic sulfur.
 (C) At temperatures above 119°C , monoclinic sulfur cannot exist at equilibrium.
 (D) The conversion of rhombic sulfur to monoclinic sulfur is endothermic.
36. Aluminum crystallizes in a face-centered cubic unit cell with an edge length of 405 pm. What is its density?

- (A) 0.674 g cm^{-3} (B) 1.35 g cm^{-3}
 (C) 2.70 g cm^{-3} (D) 5.40 g cm^{-3}

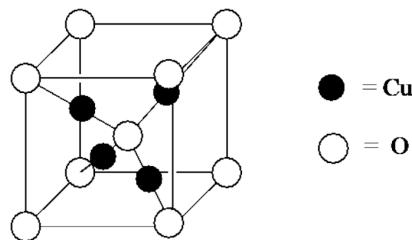
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37. The diagram is a microscopic view of a snapshot of a substance at equilibrium, with the circles representing molecules and the arrows the molecules' velocities. What state of matter is depicted?

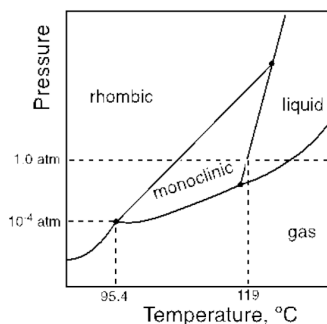


- (A) Solid (B) Liquid (C) Gas (D) Plasma
38. The vapor density of which fluorocarbon is 6.17 g L^{-1} at 23°C and 1.00 atm ?
 (A) C_3F_6 (B) C_3F_8 (C) C_4F_6 (D) C_4F_8
39. What is the principal intermolecular force that must be overcome when *n*-hexane (C_6H_{14} , bp = 69°C) is vaporized?
 (A) Hydrogen bonding
 (B) Covalent bonding between carbon atoms
 (C) Dipole-dipole forces
 (D) London dispersion forces
40. The melting point of water decreases with increasing pressure. Which is the best explanation for this observation?
 (A) Liquid water is denser than solid water at 0°C .
 (B) Melting of ice is endothermic at 0°C .
 (C) The vapor pressure of liquid water is lower than the vapor pressure of solid water at 0°C .
 (D) Solid and liquid water cannot coexist at equilibrium at 0°C at pressures different from 1 atm .
41. A cylinder containing a mixture of CO and CO_2 has a pressure of 2.00 atm at 93°C (366 K). The cylinder is then cooled to -90°C (183 K), where CO is still a gas but CO_2 is a solid with a vapor pressure of 0.25 atm . The pressure in the cylinder at this temperature is 0.90 atm . What is the mole fraction of CO_2 in the cylinder?
 (A) 0.10
 (B) 0.28
 (C) 0.35
 (D) It cannot be determined from the information given.

42. A mineral containing only copper and oxygen adopts the cubic unit cell shown below. What is the formula of this mineral?



- (A) Cu_2O (B) CuO (C) Cu_3O_2 (D) Cu_4O_9
43. Which gas has an effusion rate closest to half that of oxygen gas at a given temperature?
 (A) H_2 (B) He (C) SO_2 (D) Xe
44. Which compound has the highest normal boiling point?
 (A) 1-butanol, $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{OH}$
 (B) 2-butanol, $\text{CH}_3\text{CH}_2\text{CH}(\text{CH}_3)\text{OH}$
 (C) 2-methyl-1-propanol, $(\text{CH}_3)_2\text{CHCH}_2\text{OH}$
 (D) 2-methyl-2-propanol, $(\text{CH}_3)_3\text{COH}$
45. A portion of the phase diagram of elemental sulfur is shown below. Which statement about sulfur is correct?



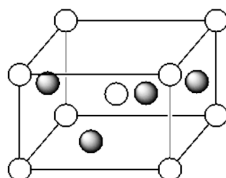
- (A) Rhombic sulfur cannot be sublimed without first converting to monoclinic sulfur.
 (B) The conversion of rhombic sulfur to monoclinic sulfur is exothermic.
 (C) Rhombic sulfur is denser than monoclinic sulfur.
 (D) At atmospheric pressure, monoclinic sulfur cannot be in equilibrium with liquid sulfur.
46. In which are the ionic compounds listed in order of increasing magnitude of their lattice energy?
 (A) $\text{CaO} < \text{RbI} < \text{LiF} < \text{LiI}$
 (B) $\text{RbI} < \text{CaO} < \text{LiI} < \text{LiF}$
 (C) $\text{LiI} < \text{LiF} < \text{CaO} < \text{RbI}$
 (D) $\text{RbI} < \text{LiI} < \text{LiF} < \text{CaO}$

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47. Sodium halides crystallize in the rock salt structure, which has a cubic unit cell with four formula units per unit cell. A sodium halide has a density of 2.83 g cm^{-3} and a unit cell edge length of 462 pm. Which compound is it?

(A) NaF (B) NaCl (C) NaBr (D) NaI

48. The tetragonal unit cell of white tin is shown below. Each circle represents a tin atom, with the gray circles sitting on the four vertical faces of the unit cell. How many tin atoms are contained in this unit cell?



(A) 2 (B) 4 (C) 8 (D) 13

49. Which gas at 600 K has the same effusion rate as methane (CH_4) at 150 K?

(A) He (B) O_2 (C) SO_2 (D) N_2O_4

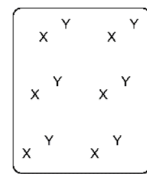
50. Ammonia (NH_3) has a higher normal boiling point (-33°C) than its heavier congeners PH_3 (bp -88°C) or AsH_3 (bp -63°C). Which is the best explanation for this difference?

(A) NH_3 is trigonal pyramidal and polar while PH_3 and AsH_3 are trigonal planar and nonpolar.
 (B) NH_3 is much more acidic than PH_3 or AsH_3 .
 (C) $\text{NH}_3(l)$ experiences stronger London dispersion forces than $\text{PH}_3(l)$ or $\text{AsH}_3(l)$.
 (D) $\text{NH}_3(l)$ has extensive hydrogen bonding while $\text{PH}_3(l)$ and $\text{AsH}_3(l)$ do not.

51. In which are the ionic solids ranked in order of increasing melting point?

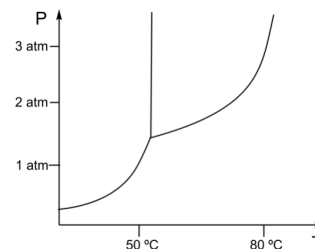
(A) $\text{KBr} < \text{NaCl} < \text{NaF} < \text{MgO}$
 (B) $\text{NaF} < \text{NaCl} < \text{MgO} < \text{KBr}$
 (C) $\text{KBr} < \text{NaCl} < \text{MgO} < \text{NaF}$
 (D) $\text{MgO} < \text{NaF} < \text{KBr} < \text{NaCl}$

52. A two-dimensional slice through the lattice of a crystalline solid containing two different elements X and Y is shown schematically and to scale below. What type of solid is it?



(A) A metallic alloy such as FeCr
 (B) A molecular solid such as IBr
 (C) An ionic compound such as LiCl
 (D) A network covalent solid such as SiC

53. A portion of the phase diagram of uranium hexafluoride (UF_6) is shown below.

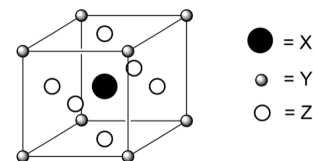


Which statements are correct?

- I. UF_6 sublimates at atmospheric pressure.
 II. At 80°C and 1.5 atm, only $\text{UF}_6(g)$ is present at equilibrium.

(A) I only (B) II only
 (C) Both I and II (D) Neither I nor II

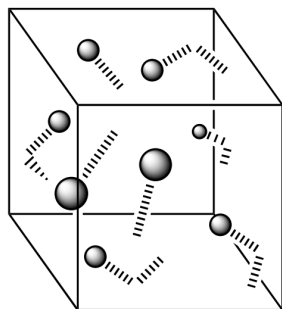
54. One unit cell of a crystal containing elements X, Y, and Z is shown below. What is its formula?



(A) XYZ (B) XYZ_3 (C) XY_4Z_2 (D) XY_8Z_6

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55. What state of matter corresponds to the diagram below?



- (A) Gas (B) Liquid
(C) Amorphous solid (D) Crystalline solid
56. The normal boiling point of 2-propanol, $(\text{CH}_3)_2\text{CHOH}$, is 83°C , while that of acetone, $(\text{CH}_3)_2\text{C}=\text{O}$, is 56°C . What is the principal reason for the greater boiling point of 2-propanol?
- (A) The O–H bond in 2-propanol is stronger than the C–H bonds in acetone.
(B) 2-Propanol experiences greater London dispersion forces than acetone.
(C) 2-Propanol experiences stronger dipole-dipole interactions than acetone.
(D) 2-Propanol experiences stronger hydrogen bonding than acetone.
57. Consider the four gases CO_2 , N_2 , CCl_4 , and He. Which is the correct order of increasing average molecular speed at 100°C ?
- (A) $\text{He} < \text{N}_2 < \text{CO}_2 < \text{CCl}_4$
(B) $\text{CCl}_4 < \text{CO}_2 < \text{N}_2 < \text{He}$
(C) $\text{He} < \text{CO}_2 < \text{N}_2 < \text{CCl}_4$
(D) $\text{CCl}_4 < \text{N}_2 < \text{CO}_2 < \text{He}$
58. Under what conditions will the behavior of a real gas best approximate the behavior of an ideal gas?
- I. High temperature II. High pressure
- (A) I only (B) II only
(C) Both I and II (D) Neither I nor II
59. A solid has a melting point of 1710°C , is soluble in water, and does not conduct electricity in the solid state. What is the most likely nature of the bonding in this solid?
- (A) Molecular covalent (B) Network covalent
(C) Ionic (D) Metallic
60. Toluene, C_7H_8 , has both a higher vapor pressure than water at 25°C and a higher normal boiling point. Which statement best explains these observations?
- (A) Liquids with higher vapor pressures typically have higher boiling points.
(B) Toluene has a higher molar mass than water.
(C) Toluene has a lower heat of vaporization than water.
(D) The density of toluene vapor is greater than that of water vapor.
61. A sample of neon gas is held at 25.0°C and 1.0 atm in a cylinder with a movable piston. Under these conditions the gas occupies 5.0 L . What volume does the gas occupy at 12.5°C and 1.0 atm ?
- (A) 2.5 L (B) 4.8 L (C) 5.2 L (D) 10 L
62. The normal boiling points of molecular fluorine, chlorine, bromine, and iodine increase in that order. Which of the following statements accounts for this increase?
- (A) The chemical reactivity decreases in that order.
(B) The London dispersion forces increase in that order.
(C) The dipole-dipole forces increase in that order.
(D) The hydrogen bonding increases in that order.
63. Quartz, SiO_2 , is the most common mineral found on the surface of the earth. What is the best explanation for the fact that quartz is hard and has a high melting point?
- (A) Quartz crystals are extended structures in which each atom forms strong covalent bonds with all of its neighboring atoms.
(B) Quartz crystals consist of positive and negative ions that are attracted to one another.
(C) Quartz crystals are formed under extremes of temperature and pressure.
(D) Silicon and oxygen atoms are especially hard because of their electronic structure.
64. The critical point of carbon dioxide is 304 K and 73 atm . Under which conditions is carbon dioxide a liquid?
- I. 303 K and 73 atm II. 305 K and 74 atm
- (A) I only (B) II only
(C) Both I and II (D) Neither I nor II

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65. The lattice energy (energy required to separate the ions in an ionic solid) of MgO is much larger than that of LiF. What contributes the most to this difference?

- (A) Mg^{2+} is a smaller ion than Li^+ , and O^{2-} is a smaller ion than F^- .
- (B) F is more electronegative than O, and Li is more electropositive than Mg.
- (C) MgO contains doubly charged ions, while LiF contains singly charged ions.
- (D) MgO contains more electrons than LiF.

66. Which best describes the bonding in Cu(s)?

- (A) The copper atoms are positively charged in a sea of delocalized electrons.
- (B) The copper atoms are alternately positively and negatively charged.
- (C) The copper atoms form covalent bonds to adjacent copper atoms.
- (D) The copper atoms form hydrogen bonds to adjacent copper atoms.

67. A sample of O_2 gas at a given temperature and pressure has a density of 1.30 g L^{-1} . What is the density of gaseous propane (C_3H_8) under the same conditions?

- (A) 1.30 g L^{-1} (B) 1.79 g L^{-1}
- (C) 1.96 g L^{-1} (D) 2.60 g L^{-1}

68. Which statements about the behavior of gaseous H_2 molecules in a container at 1 atm and 298 K are correct?

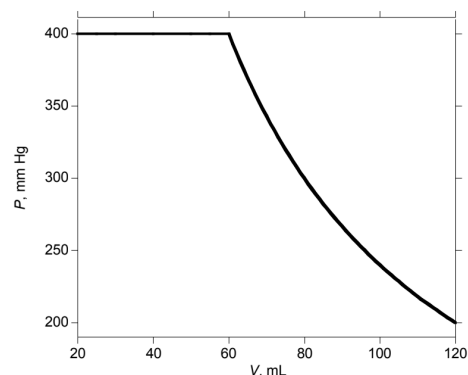
- I. All H_2 molecules are moving at the same speed.
- II. The H_2 molecules are colliding more frequently with the walls of the container than they would in the same container at 398 K.

- (A) I only (B) II only
- (C) Both I and II (D) Neither I nor II

69. Which species has a normal boiling point closest to the normal boiling point of argon, Ar?

- (A) H_2 (B) N_2 (C) F_2 (D) Cl_2

70. A sample of methanol, CH_3OH , is introduced into an evacuated chamber with a movable piston. The pressure is measured as a function of the volume of the container while the temperature is maintained at 50°C , and the graph below is obtained.



Which statements are correct?

- I. At volumes less than 60 mL, only liquid methanol is present.
- II. At volumes greater than 60 mL, only gaseous methanol is present.

- (A) I only (B) II only
- (C) Both I and II (D) Neither I nor II

71. How many nearest neighbors does each silicon atom have in solid Si?

- (A) 4 (B) 6 (C) 8 (D) 12

72. Which sample contains the smallest number of atoms?

- (A) 1.0 L of Ar at STP
- (B) 1.0 L of H_2 at STP
- (C) 1.0 L of Ar at 25°C and 760 mm Hg
- (D) 1.0 L of H_2 at 0°C and 900 mm Hg